How Do LSTMs and Attention Improve on Vanilla RNNs for Medical Speciality Classification

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1 Introduction

In this project, we investigate whether LSTM gating and attention pooling improve classification of clinical records into 12 medical specialities compared with vanilla RNNs. We compare four bidirectional models, combining RNNs and LSTMs with and without attention. Since keywords often explicitly reveal the correct speciality, we also test different input-field combinations and keyword-prefix removal to distinguish architectural improvements from reliance on label cues.

2 Data

2.1 Data Analysis

The data is strongly imbalanced (Surgery, for example, has about 30x more representation than Dermatology).

Firstly, in 82.9% of the training records, the first keyword is a speciality name and matches the label, which serves as a baseline of what performance we look to achieve. Later on we noticed that removing this first keyword from the dataset causes accuracy to decrease from about 0.9 to 0.4.

We also noticed that there are 170 duplicate records (1.0 cosine similarity) that yet have 2 different labels. Common conflicting pairs include Surgery and Orthopedic, Cardiovascular-Pulmonary and Gastroenterology. This creates ambiguity that cannot be solved from the input text alone and bounds the achievable performance of our models.

Initially we tried to reduce description size, but we found that the most frequent eight-word phrases were strongly associated with specific specialities (which we measured through KL divergence between class distribution and the label distribution of these phrases). However, as we will see, the best performance was obtained without the use of the description feature.

2.2 Data Augmentation

We used word dropout and LLM (GPT 6) aided paraphrasing on all features used, applied to the training partition after the split, avoiding both noise at the character level and removal of negations. The dataset size increased by 21% and got more balanced (Shannon Entropy went from 3.041 to 3.123, where the maximum for 12 classes is 3.585) To prevent data leakage during evaluation, synthetic examples were associated with their source records and restricted to the training partition of each cross-validation fold.

3 Models

We adopted Multinomial Naive Bayes with TF-IDF features as baseline. To isolate the contributions of LSTM gating and attention, we designed a controlled comparison of four bidirectional models: vanilla RNN (BiRNN) and LSTM (BiLSTM), each

with either Masked Mean Pooling or Single-Head Additive Attention Pooling. Keeping the remaining architecture and training settings fixed allows us to evaluate each component independently.

All four models share trainable word embeddings, a bidirectional recurrent encoder, a pooling layer and a linear classifier predicting one of 12 specialities. Bidirectionality captures context from both directions. Vanilla RNNs update hidden states recurrently [1], but suffer from vanishing gradients [2], limiting long-range dependencies. LSTMs address this using memory cells controlled by input, forget and output gates [3, 4], potentially benefiting lengthy clinical transcriptions.

Masked Mean Pooling averages valid token representations equally. In contrast, Additive Attention Pooling [5] learns a score for each hidden state, applies Softmax to obtain attention weights and computes a weighted sum. This allows the model to emphasize informative passages rather than treating all tokens equally.

Beyond comparing architectures, we tested input-field combinations, removed label-revealing keyword prefixes and explored class weighting and data augmentation. These experiments help distinguish architectural improvements from gains caused by input selection or class imbalance. Shared preprocessing, hyperparameters and evaluation procedures are detailed in Section 4.

4 Experimental Setup

4.1 Parameters/Hyperparameters

To isolate the effects of LSTM gating and Attention, we kept hyperparameters fixed across all four recurrent models: 128-dimensional word embeddings, 128 hidden units per direction, one recurrent layer and a dropout rate of 0.3. Models were trained using AdamW [6], with a learning rate of 0.001, batch size of 16 and a maximum of 30 epochs. We used cross-entropy loss with optional class weighting to address class imbalance, and early stopping based on validation macro-F1 with patience of 6 epochs.

4.2 Data Preparation and Input Configurations

For the neural models, selected input fields were put together into a single text, converted to lowercase and split into tokens using a regular expression. We used a vocabulary of at most 20 000 tokens, built from the training data, excluding those appearing fewer than twice. We evaluated individual fields and combinations of them to study their influence on classification. We also tested what happens when the speciality name is removed from the beginning of the keywords.

4.3 Data Splits and Evaluation Metrics

We evaluated 2,613 valid labeled records using five-fold cross-validation, preserving class distributions and grouping identical transcriptions to prevent data leakage. In each fold, approximately 64% of records were used for training, 16% for validation and early stopping and 20% for testing, with identical splits across all models for fair comparison. We measured accuracy, macro-F1 and minimum per-class F1 to assess both overall performance and minority-class classification.

5 Results

Table 1 presents the five-fold cross-validation results using keywords, sample names and transcriptions, our best-performing input combination. BiLSTM with attention achieved the highest accuracy (91.70%) and macro-F1 (89.23%), while BiRNN with attention achieved the highest mean minimum-class F1 (0.766). Both attention models met the evaluation thresholds in every fold.

For BiLSTM with attention, using all four fields or only keywords and sample names resulted in

Table 1: Five-fold mean scores using keywords, sample names and transcriptions.

Model	Acc.	Macro-F1	Min F1
Naive Bayes	0.413	0.145	0.000
BiRNN	0.719	0.616	0.055
BiLSTM	0.876	0.810	0.392
BiRNN + Att.	0.910	0.891	**0.766**
BiLSTM + Att.	**0.917**	**0.892**	0.746

slightly lower accuracy (90.89% in both cases). The best model's pooled per-class F1 ranged from 0.753 for Psychiatry-Psychology to 0.978 for Ophthalmology. Complete per-class results are provided in Appendix A, together with the confusion matrix of our best model (Figure 1).

Tables 2 and 3 further compare input configurations, augmentation and class weighting.

6 Discussion

Our results suggest that attention plays a particularly important role in this classification task . Without attention, BiLSTM outperformed BiRNN by 15.69 percentage points in accuracy, whereas with attention, this difference decreased to only 0.65 points. A possible explanation is that attention allows models to emphasize informative parts of lengthy clinical records, rather than treating all hidden states equally. This may compensate for some limitations of vanilla RNNs, although it does not replace the memory mechanisms of LSTMs.

Input selection proved decisive. In 82.9% of training records, the first keyword matched the correct speciality. Removing this prefix reduced BiLSTM with attention accuracy from 86.07% to 42.75% in the keywords-only experiment. Thus, high performance partly reflects explicit label cues rather than clinical text understanding.

As shown in Figure 1, frequent misclassifications include Cardiovascular-Pulmonary as General Medicine (19 cases), Cardiovascular-Pulmonary as Surgery (13 cases), and Neurology as General Medicine (12 cases). The following examples from the separately exported validation predictions illustrate possible causes:

1. Correct: Cardiovascular-Pulmonary; Predicted: General Medicine. "Pneumonia, failed outpatient treatment ... Hypoxia ... Rheumatoid arthritis." The combination of pulmonary symptoms and general medical conditions may have encouraged the broader classification. General Medicine as a label, however, risks overlap with other classes.
2. Correct: Cardiovascular-Pulmonary; Predicted: Surgery."PREOPERATIVE DIAGNOSIS: Symptomatic pericardial effusion ... PROCEDURE PERFORMED: Subxiphoid pericardiotomy." Surgical terminology may have outweighed the cardiovascular context (noting that duplicate datapoints occur and surgery is prominent in these ambiguous cases).
3. Correct: Cardiovascular-Pulmonary; Predicted: Neurology."EXAMINATION: Cardiac catheterization ... INDICATION: Syncope with severe aortic stenosis." Syncope is also associated with neurological evaluation, potentially introducing ambiguity despite the cardiac procedure.

Overall, attention reduced the performance gap between BiRNN and BiLSTM, directly addressing our research question. Nevertheless, overlapping medical terminology, conflicting labels and reliance on explicit speciality keywords remain important limitations.

7 Future Work

Future work should examine whether attention maintains its advantage without label-revealing keyword prefixes and explore pretrained medical embeddings to better distinguish overlapping specialities. This would help assess whether attention captures meaningful clinical information beyond explicit label cues.

References

[1] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

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[3] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, November 1997.

[4] Felix A. Gers, Jürgen Schmidhuber, and Fred Cummins. Learning to forget: Continual prediction with lstm. *Neural Computation*, 12(10):2451–2471, October 2000.

[5] Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, and Eduard Hovy. Hierarchical attention networks for document classification. In *Proceedings of NAACL-HLT 2016*, pages 1480–1489. Association for Computational Linguistics, June 2016.

[6] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*. OpenReview.net, May 2019.

LLMs were used to assist with code development, debugging and improving the English in this report. GPT-6 Sol High Reasoning was also used to generate paraphrases of clinical descriptions for data augmentation.

Appendix A: Extra Figures and Tables

Maximum one page (it does not count for the 3 pages limit): just extra figures and tables. **Notice that we should be able to understand the 3 pages paper without these figures and tables.**

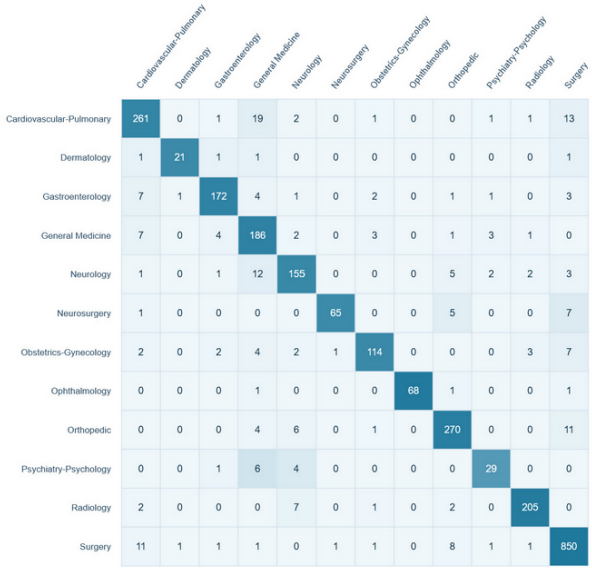


Figure 1: Pooled confusion matrix of BiLSTM with Attention using keywords, sample names and transcriptions. Rows represent correct specialities and columns represent predictions.

Table 2: No Weighted (Min Class F1 ± SD)

	Baseline	BR	BR-A	BL	BL-A
A4F	0	0.173±0.158	0.727±0.114	0.296±0.196	0.685±0.105
Key only	0	0.408±0.142	0.455±0.175	0.450±0.167	0.450±0.167
Key(-pre)	0	0.000±0.000	0.000±0.000	0.036±0.081	0.061±0.090
Key+S+T	0	0.055±0.122	0.766±0.055	0.392±0.252	0.746±0.082
Key+Samp	0	0.644±0.077	0.746±0.085	0.741±0.068	0.743±0.078

**Baseline=Naive Bayes, BR=BiRNN, BL=BiLSTM, -A=Attention.*

Table 3: BiLSTM with Attention (Class Weighting vs Normal)

	BL-A (CW)	BL-A
KEY + SAMP	0.752 ± 0.071	0.743 ± 0.078
AUG. RWD	0.716 ± 0.084	0.753 ± 0.075
AUG. LLM	0.746 ± 0.062	0.737 ± 0.088
AUG. RWD+LLM	0.713 ± 0.091	0.676 ± 0.102

**CW=Class Weight (sqrt_inv), BL=BiLSTM, -A=Attention. Values represent fold mean Min Class F1 ± SD.*